DATA ARCHITECTURE CASE STUDIES · NO. 1

Detecting Inserts, Updates, and Deletes Without Tombstones

Hash-Diff Synchronization over a Linked Server

How a nightly replica of a vendor ERP stays exact under a hard query timeout, a query-text limit, and no tombstones: the method, shown in the SQL it generates.

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I designed and built a nightly sync that kept a warehouse copy of a vendor ERP exact: roughly 700 tables and 2 billion rows, reconciled every night in about 38 minutes, replacing a process that took about 9 hours. The client's code stays with the client, so I rebuilt the method from scratch against a generated stand-in ERP, in the same T-SQL, over a real SQL Server linked server. The generated statements below are real output from that rebuild; the few code excerpts are marked where abridged.

The constraints decide the design

The ERP was reachable only through a linked server backed by the vendor's proprietary driver. Six facts shaped everything that follows:

- **A 15-minute hard timeout** on every query.
- **A limit on query-text length**, which caps how much any one statement can say.
- **Read-only access.** No staging tables, temp tables, or views on the vendor's side.
- **No tombstones.** A deleted row simply disappears.
- **Create and modify dates that cannot be trusted**: sometimes missing, sometimes not updated when a row changes.
- **Undeclared keys.** The real keys are composite, an int plus codes and sometimes a date, and live only in the vendor's data dictionary.

Untrustworthy dates rule out filtering for recent changes, and missing tombstones rule out asking for deletes. What remains is comparing every row, every night, and making that cheap.

One path for every table

One procedure runs once per table, many tables in parallel. The first load, a nightly run, and a full rebuild all take the same eight steps (a rebuild skips the guard and treats every key as new), and every statement sent to the ERP is generated from two things: a copy of the ERP's column metadata, and a hand-kept manifest of each table's key columns.

1	2	3	4	5	6	7	8
Pull every key and a row hash	**Guard** against a short pull	**Compare** new, changed, deleted	**Prune** to a request list	**Fetch** keys sent in the query text	**Apply** in one transaction	**Reconcile** every count, or roll back	**Log** every count, then drop staging

Three staging tables per table

Every step reads or writes three tables in a `stage` schema on the warehouse, created fresh at the start of each table's run and named for the table they serve. They are ordinary tables, not `#temp` tables, so a table that fails leaves its last attempt's staging in place to inspect. Because the names belong to the table rather than the run, two syncs of the same table would collide, so each table's run holds an application lock.

TABLE	HOLDS	WRITTEN	READ
`__hashkey`	Every pulled key and its row hash, then working columns: the action (new or changed), the key as literal text, and its batch number	1 (pull), 3 (action), 4 (prune), 5 (literals, batches)	3 (compare), 5 (fetch), 6 (apply)
`__deleted`	Keys active in the replica but missing from the pull	3 (compare)	3 (count), 6 (apply)
`__rows`	The full rows fetched from the ERP	5 (fetch)	5 (count), 6 (apply), joined to `__hashkey` for the hash

```
CREATE TABLE stage.[ar_dunning__hashkey] (
    [dunning_id] int NOT NULL, [dunning_code] varchar(12) NOT NULL,
    [line_code] varchar(8) NOT NULL,
    _row_hash char(32) NOT NULL, _action char(1) NULL,
    _key_literal nvarchar(max) NULL, _batch int NULL);
CREATE CLUSTERED INDEX cx ON stage.[ar_dunning__hashkey]
    ([dunning_id], [dunning_code], [line_code]);

CREATE TABLE stage.[ar_dunning__deleted] (
    [dunning_id] int NOT NULL, [dunning_code] varchar(12) NOT NULL,
    [line_code] varchar(8) NOT NULL);

CREATE TABLE stage.[ar_dunning__rows] (
    [dunning_id] int NULL, [dunning_code] varchar(12) NULL,
    [line_code] varchar(8) NULL, [created_on] datetime NULL, …);
    /* all thirteen columns, each nullable */
```

Generated per table: the key columns come from the manifest, typed from the ERP's metadata, and `__rows` mirrors the ERP's columns. Reformatted and abridged from the one statement actually run. The clustered index on the key serves the compare and the apply.

1 Pull every key and a hash of its whole row

One query per table returns each key and an MD5 of the entire row, computed on the ERP's side, so only keys and 32-character hashes cross the wire. In production this single pull finished inside the timeout even at 500 million rows. For a thirteen-column table it looks like this, reformatted from the one line actually sent:

```
DECLARE @d varchar(3) = '$|$', @n char(1) = CHAR(0), @m varchar(max) = '';
SELECT [dunning_id], [dunning_code], [line_code],
       CONVERT(char(32), HASHBYTES('MD5', CONCAT(@m, CONCAT(@m,
           @d, ISNULL(CONVERT(varchar(20), [dunning_id]), @n),
           @d, ISNULL([dunning_code], @n),
           @d, ISNULL([line_code], @n),
           @d, ISNULL(CONVERT(varchar(27), [created_on], 121), @n),
           @d, ISNULL(CONVERT(varchar(27), [modified_on], 121), @n),
           @d, ISNULL(CONVERT(varchar(40), [hours_001]), @n),
           @d, ISNULL([label_002], @n),
           @d, ISNULL(CONVERT(varchar(20), [ext_id_003]), @n),
           @d, ISNULL(CONVERT(varchar(20), [closed_004]), @n),
           @d, ISNULL(CONVERT(varchar(40), [rate_005]), @n),
           @d, ISNULL(CONVERT(char(10), [start_date_006], 23), @n),
           @d, ISNULL([note_007], @n),
           @d, ISNULL(CONVERT(varchar(40), [balance_008]), @n)))), 2)
FROM dbo.[ar_dunning]
```

742 characters for thirteen columns. The outer `CONCAT` holds one inner `CONCAT` per 126 columns, so a narrow table has exactly one. The widest stand-in table, 318 columns, needs about 12,800 characters for the same statement.

Every detail in that statement is there to make the hash honest:

- **Each type gets its own cast**, in a style that keeps full precision: 2 for money, 3 for float, 121 for datetime, 23 for date. The defaults round money to two decimals, cut float to six significant digits, and drop datetime's seconds.
- **NULL becomes `CHAR(0)`**, a character real text values do not contain, so NULL and an empty string hash differently.

- **The delimiter is `$|$`**, so a pipe moving from one value to its neighbor cannot produce the same string.
- **Every `CONCAT` starts from `@m`, a `varchar(max)`.** Without it, SQL Server silently truncates at 8,000 characters, or 4,000 once an `nvarchar` column joins in. I hit this in production; the rebuild hit it again independently. `CONCAT` takes at most 254 arguments, so wide tables nest one per 126 columns.

Each row below is a one-row change made in the stand-in ERP. An obvious first-draft hash (default conversions, NULL as an empty string, a single pipe, no `varchar(max)`) misses every one; the sync catches them all.

CHANGE	BEFORE → AFTER	FIRST-DRAFT HASH	THIS SYNC
NULL becomes an empty string	`NULL` → `''`	misses it	changed
money, fourth decimal	27563.3800 → 27563.3801	misses it	changed
float, 13th significant digit	1.7917896271428570e+008 → 1.7917896271446490e+008	misses it	changed
datetime, 10 milliseconds	04:37:28.047 → 04:37:28.057	misses it	changed
a pipe moves between neighbors	`'ab'`, `'|cd'` → `'ab|'`, `'cd'`	misses it	changed
a change past the 4,000th character	…xxx → …xxy	misses it	changed

The ERP's hash is stored on each replica row, and the next night compares against it. The warehouse never reproduces the ERP's casting, so the two sides never hash the same row differently. MD5 is enough here: it detects change rather than tampering, and the odds of two versions of a row colliding are negligible.

2–4 Guard, compare, prune

A short pull looks exactly like mass deletion, so before comparing anything the procedure checks the count. The replica may shrink by deletes, but only so far. `@wipe` marks a full rebuild, and `@allowance` is each table's permitted shrink, 1 percent by default:

```
IF @wipe = 0 AND @active_before > 0
   AND @pulled < @active_before * (1 - @allowance)
    THROW 50010, @msg, 1;  -- logged, alerted, and not retried
```

Then one pass over the pulled hashes classifies every key. A key the replica lacks is new; a key whose hash differs, or that was deleted and has come back, is changed; a key active in the replica but absent from the pull was deleted in the ERP. Unchanged keys are dropped, and what remains is the request list.

```
UPDATE h SET _action =
    CASE WHEN t.[dunning_id] IS NULL THEN 'N'
         WHEN t._is_deleted = 1 OR t._row_hash <> h._row_hash THEN 'C' END
FROM stage.[ar_dunning__hashkey] h
LEFT JOIN replica.[ar_dunning] t
  ON h.[dunning_id] = t.[dunning_id] AND h.[dunning_code] = t.[dunning_code]
 AND h.[line_code] = t.[line_code];

INSERT INTO stage.[ar_dunning__deleted]
       ([dunning_id], [dunning_code], [line_code])
SELECT t.[dunning_id], t.[dunning_code], t.[line_code]
FROM replica.[ar_dunning] t
WHERE t._is_deleted = 0
  AND NOT EXISTS (SELECT 1 FROM stage.[ar_dunning__hashkey] h
                  WHERE h.[dunning_id] = t.[dunning_id]
                    AND h.[dunning_code] = t.[dunning_code]
                    AND h.[line_code] = t.[line_code]);
```

5 Send the keys back inside the query text

Nothing can be created on the ERP, so there is nowhere to put a list of keys except the statement itself. Each requested key becomes a literal tuple in a `VALUES` list, joined on every key column, and each batch's rows land in `__rows` through `INSERT … EXEC` over the linked server:

```
SELECT s.[dunning_id], s.[dunning_code], s.[line_code], s.[created_on],
       s.[modified_on], s.[hours_001], s.[label_002], s.[ext_id_003],
       s.[closed_004], s.[rate_005], s.[start_date_006], s.[note_007], s.[balance_008]
FROM dbo.[ar_dunning] s
JOIN (VALUES (1,'C''3','L000'),
             (1,'C''3','L001'),
             (1,'C''3','L002')) v([dunning_id], [dunning_code], [line_code])
  ON s.[dunning_id] = v.[dunning_id]
 AND s.[dunning_code] = v.[dunning_code]
 AND s.[line_code] = v.[line_code]
```

Three keys shown; a first-load batch holds hundreds to over a thousand. The code `C'3` goes in as `'C''3'`. Unescaped, the first quoted code would end the string and break the batch.

The batch size is arithmetic on the statement. The select list is the same for every batch of a table, so what remains of the text limit is the room for keys. A running sum of literal lengths closes each batch before it would overflow, with a cap of 5,000 rows:

```
WITH a AS (SELECT _batch,
                  SUM(LEN(_key_literal) + 1)
                    OVER (ORDER BY [dunning_id], [dunning_code], [line_code]
                          ROWS UNBOUNDED PRECEDING) AS running_len
           FROM stage.[ar_dunning__hashkey]),
     b AS (SELECT _batch, running_len,
                  (running_len - 1) / (@room - @longest_literal) AS char_batch
           FROM a),
     c AS (SELECT _batch, char_batch,
                  ROW_NUMBER() OVER (PARTITION BY char_batch
                                     ORDER BY running_len) AS rn
           FROM b),
     d AS (SELECT _batch,
                  DENSE_RANK() OVER (ORDER BY char_batch, (rn - 1) / @row_cap) AS nb
           FROM c)
UPDATE d SET _batch = nb;
```

From the procedure, with variables renamed for reading. `@room` is the text limit minus the statement's fixed text. Dividing by `@room - @longest_literal` keeps one longest key's worth of space spare, so a batch never overflows. `@row_cap` is 5,000. On the first load, batches landed at up to 32,764 characters against the stand-in's 32,767 limit. A 318-column table fits about 2,200 keys per batch; a narrow one about 2,550.

6–7 Apply everything at once, then prove the counts

Fetched rows land in staging. Nothing touches the replica until every batch is in; then one transaction flags the deletes, removes the changed rows, and inserts the fetched ones, each stamped with its load time and the hash from the pull. Each statement captures its own row count, and before the transaction commits, step 7 checks each one against what the earlier steps said to expect: rows flagged against keys found deleted, rows removed against keys found changed, rows inserted against keys requested, and the replica's active rows against keys pulled. Any mismatch rolls the whole table back:

```
BEGIN TRAN;
  UPDATE t SET _is_deleted = 1
  FROM replica.[ar_dunning] t JOIN stage.[ar_dunning__deleted] d ON …;
  SET @flagged = @@ROWCOUNT;

  DELETE t
  FROM replica.[ar_dunning] t JOIN stage.[ar_dunning__hashkey] h ON …
  WHERE h._action = 'C';
```

```
  SET @removed = @@ROWCOUNT;

  INSERT INTO replica.[ar_dunning]
         (/* 13 columns */, _row_hash, _loaded_at, _is_deleted)
  SELECT /* 13 columns */, h._row_hash, @run_at, 0
  FROM stage.[ar_dunning__rows] r JOIN stage.[ar_dunning__hashkey] h ON …;
  SET @inserted = @@ROWCOUNT;

  SELECT @active_after = COUNT(*)
  FROM replica.[ar_dunning] WHERE _is_deleted = 0;

  -- Expected, from steps 1 to 4: @deleted keys missing from the pull,
  -- @changed keys whose hash moved, @requested keys fetched (new plus changed),
  -- @pulled keys pulled.
  IF @flagged <> @deleted OR @removed <> @changed
     OR @inserted <> @requested OR @active_after <> @pulled
      THROW 50013, @msg, 1;  -- rolls back: the table stays exactly as it was
COMMIT;
```

Abridged: the join conditions are the same three-column key match as above. The apply runs as one generated statement that returns its counts through output parameters; they are shown inline here.

That makes failure cheap. Retries are nested, three per batch inside the procedure and three per table in the runner, and a failed `INSERT … EXEC` inserts nothing, so a retry never duplicates a row. A table that fails every attempt is left exactly as it was the night before, and the next run picks up everything it missed. The one email per bad run, as the original sync's SSIS package sent, says so:

```
Run 7 (2026-01-07 02:00:00): 119 tables succeeded, 1 did not.

pj_project: stopped by the variance guard after 1 attempt.
  Error 50010: Variance guard: pulled 129344 keys, replica has 136002 active rows, allowance 1.0 percent.
  Not retried. The replica was left untouched. Check the source before the next run.
```

From the failure proof: 5% of a table vanished from the ERP, as in a partial restore. Without the guard, those rows would have been flagged deleted.

8 Log every count, then clean up

Every table writes one log row per run: when it started and finished, how many keys it pulled, how many were new, changed, and deleted, how many rows it requested, fetched, and wrote, how many batches it sent and retried, and the longest statement it sent to the ERP. A table that fails records how far it got, from its pull, compare, and fetch counts, and the error that stopped it. The alert email above is built from this log.

Staging is dropped only after the log records success. A failed table keeps the staging from its last attempt for inspection until its next run starts fresh. Once the changes are committed, an error in logging or cleanup is recorded against the run but does not mark the table failed, because its replica is already right; only a missing log row, which leaves nothing to record against, raises the error.

```
UPDATE sync.run_table_log
SET status = 'succeeded', finished_at = SYSDATETIME(),
    pulled_keys = @pulled, new_keys = @new, changed_keys = @changed,
    deleted_keys = @deleted, fetched = @fetched, inserted = @inserted, …
WHERE log_id = @log_id;
IF @@ROWCOUNT <> 1
    THROW 50014, 'The log row for this run is missing, so staging was kept.', 1;

DROP TABLE stage.[ar_dunning__hashkey];
DROP TABLE stage.[ar_dunning__deleted];
DROP TABLE stage.[ar_dunning__rows];
```

Abridged: the log row holds eighteen counts and timings, and the staging names are generated per table.

Proving it without the client's system

The stand-in is generated in T-SQL from a seed: 120 tables and 1.2 million rows with undeclared composite keys, quoted codes, untrustworthy dates, and tables up to 318 columns, plus nightly activity that inserts, updates, deletes without a trace, and re-creates deleted keys. The read-only login is real. The timeout is real but scaled down with the data, to 30 seconds from 15 minutes, and a wrapper turns it into an error as the vendor's driver did. The text limit is the one constraint simulated, at 32,767 characters, in the single procedure every ERP call passes through.

To check the rebuild, an independent audit compares every row of every table with the ERP after each run. It is test scaffolding, not part of the sync: it reads the ERP through its own linked server and fingerprints rows with SQL Server's JSON rendering and SHA-256, so a bug in the sync's hashing cannot hide from it. The old process ran beside the new one on the same nights, rebuilt as it was: most tables reloaded in full, the large ones filtered by date and upserted, most on only part of their key.

960 of 960
table audits clean over the first load and seven nights

1,884 rows
fetched on night 7, after comparing all 1,227,989 keys

68,936 rows
wrong or missing in the rebuilt old process's copy after seven nights, with no errors raised

What I would do differently

- **Close the gap between the pull and the fetch.** The stored hash comes from the pull and the values from the fetch. A row that changes in between, then changes back before the next pull, keeps the in-between values. I never saw transactions land at that hour, but it is possible; fetching each row's hash with its values would close it.
- **Retry only what a retry can fix.** Every failure except the variance guard is retried, including errors that can never succeed on retry, such as schema drift or a statement over the text limit.
- **Handle schema changes in place.** A vendor change to a table's columns stops that table, by name, until it is rebuilt from scratch. Adding the column and refetching that one table would keep it running.

The code, the stand-in, and every proof are in data-architecture-case-studies / erp-hashdiff-sync. The statements above come from its T-SQL: sql/warehouse/30_sync_process.sql.